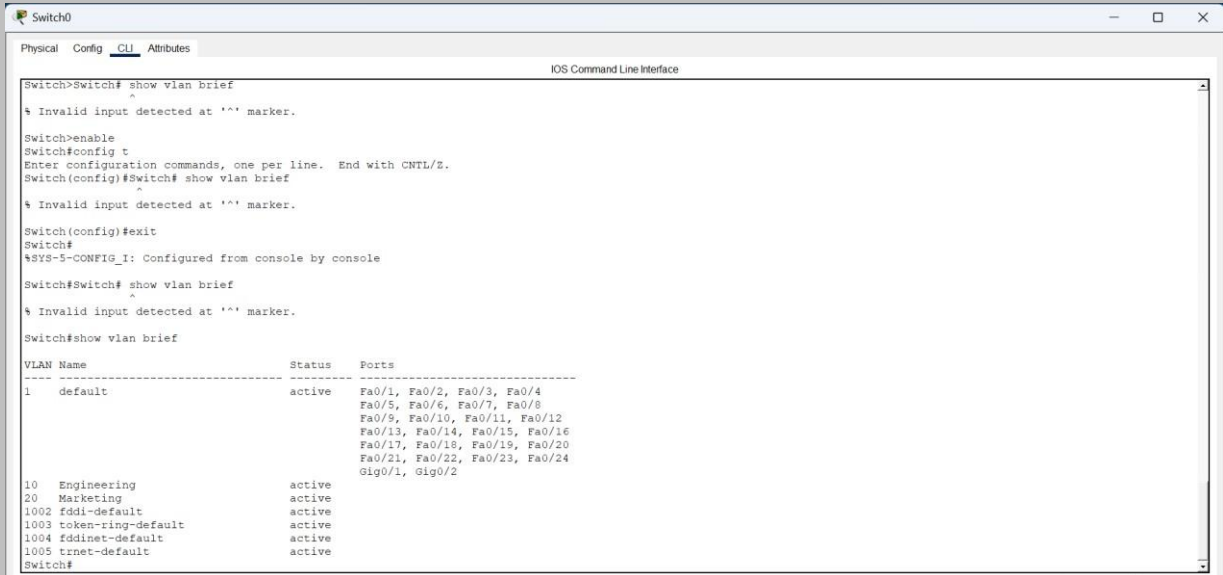


VLAN CONCEPT

1. On a Cisco Packet Tracer switch, create two VLANs named 'Engineering' (VLAN 10) and 'Marketing' (VLAN 20) using the CLI commands.



```
Switch0
Physical Config CLI Attributes
IOS Command Line Interface

Switch>Switch# show vlan brief
^
% Invalid input detected at '^' marker.

Switch>enable
Switch#config t
Enter configuration commands, one per line. End with CNTL/Z.
Switch(config)#Switch# show vlan brief
^
% Invalid input detected at '^' marker.

Switch(config)#exit
Switch#
^SYS-5-CONFIG_I: Configured from console by console

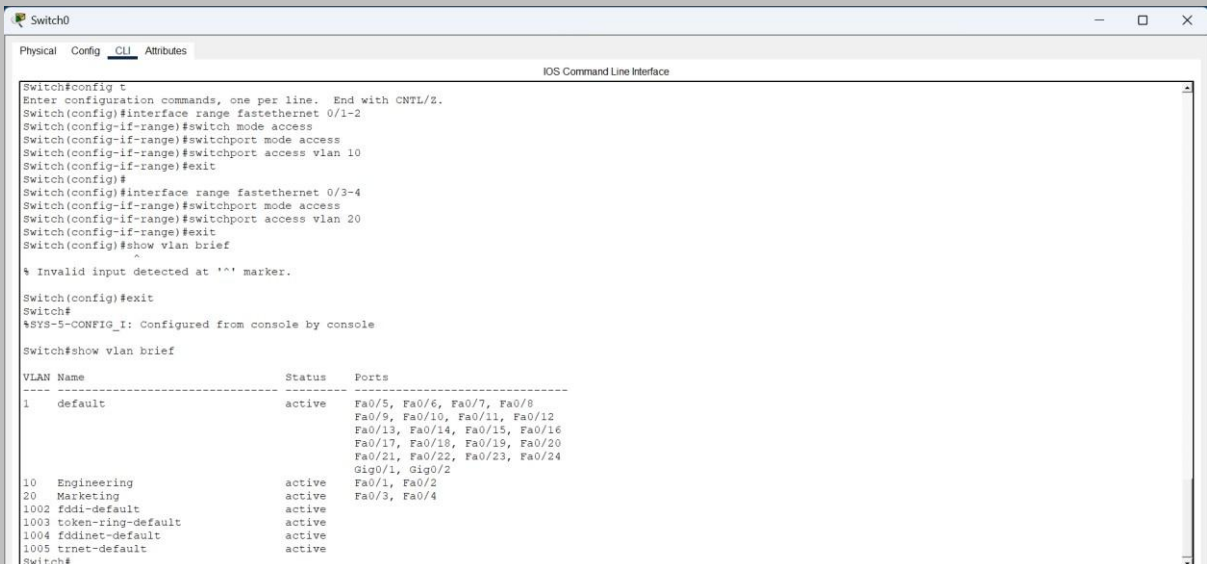
Switch>Switch# show vlan brief
^
% Invalid input detected at '^' marker.

Switch#show vlan brief

VLAN Name                Status    Ports
-----
1    default                active    Fa0/1, Fa0/2, Fa0/3, Fa0/4
                                           Fa0/5, Fa0/6, Fa0/7, Fa0/8
                                           Fa0/9, Fa0/10, Fa0/11, Fa0/12
                                           Fa0/13, Fa0/14, Fa0/15, Fa0/16
                                           Fa0/17, Fa0/18, Fa0/19, Fa0/20
                                           Fa0/21, Fa0/22, Fa0/23, Fa0/24
                                           Gig0/1, Gig0/2

10   Engineering            active
20   Marketing              active
1002 fddi-default          active
1003 token-ring-default    active
1004 fddinet-default       active
1005 trnet-default         active
Switch#
```

2. Assign ports FastEthernet0/1 and FastEthernet0/2 to VLAN 10, and ports FastEthernet0/3 and FastEthernet0/4 to VLAN 20 on your switch.



```
Switch0
Physical Config CLI Attributes
IOS Command Line Interface

Switch#config t
Enter configuration commands, one per line. End with CNTL/Z.
Switch(config)#interface range fastethernet 0/1-2
Switch(config-if-range)#switchport mode access
Switch(config-if-range)#switchport access vlan 10
Switch(config-if-range)#exit
Switch(config)#
Switch(config)#interface range fastethernet 0/3-4
Switch(config-if-range)#switchport mode access
Switch(config-if-range)#switchport access vlan 20
Switch(config-if-range)#exit
Switch(config)#show vlan brief
^
% Invalid input detected at '^' marker.

Switch(config)#exit
Switch#
^SYS-5-CONFIG_I: Configured from console by console

Switch#show vlan brief

VLAN Name                Status    Ports
-----
1    default                active    Fa0/5, Fa0/6, Fa0/7, Fa0/8
                                           Fa0/9, Fa0/10, Fa0/11, Fa0/12
                                           Fa0/13, Fa0/14, Fa0/15, Fa0/16
                                           Fa0/17, Fa0/18, Fa0/19, Fa0/20
                                           Fa0/21, Fa0/22, Fa0/23, Fa0/24
                                           Gig0/1, Gig0/2

10   Engineering            active    Fa0/1, Fa0/2
20   Marketing              active    Fa0/3, Fa0/4
1002 fddi-default          active
1003 token-ring-default    active
1004 fddinet-default       active
1005 trnet-default         active
Switch#
```

3. Write a short document (in a text file) listing the VLAN IDs, names, and the ports assigned to each VLAN from your configuration.

VLAN Configuration Details

VLAN ID: 10 VLAN Name: Foodies Assigned Ports: FastEthernet0/1 - FastEthernet0/2

VLAN ID: 20 VLAN Name: Shoppers Assigned Ports: FastEthernet0/3 - FastEthernet0/4

VLAN ID: 30 VLAN Name: Streamers Assigned Ports: FastEthernet0/5 - FastEthernet0/6

4. Explain, in your own words, two benefits of using VLANs in a college campus network that has separate labs for coding and networking.

1. **Better Network Organization:** VLANs can separate the coding lab and networking lab into different virtual networks, even if they use the same physical switches. This makes the network easier to manage.
2. **Improved Security and Performance:** Devices in the coding lab can be separated from devices in the networking lab, reducing unnecessary traffic and improving security. Problems or broadcasts in one VLAN will also have less impact on the other VLAN.